

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

Which of the following equations represents a circle in the xy -plane that intersects the y -axis at exactly one point?

Answer

- A. $(x - 8)^2 + (y - 8)^2 = 16$
- B. $(x - 8)^2 + (y - 4)^2 = 16$
- C. $(x - 4)^2 + (y - 9)^2 = 16$
- D. $x^2 + (y - 9)^2 = 16$

Correct Answer: C

Rationale

Choice C is correct. The graph of the equation $(x - h)^2 + (y - k)^2 = r^2$ in the xy -plane is a circle with center (h, k) and a radius of length r . The radius of a circle is the distance from the center of the circle to any point on the circle. If a circle in the xy -plane intersects the y -axis at exactly one point, then the perpendicular distance from the center of the circle to this point on the y -axis must be equal to the length of the circle's radius. It follows that the x -coordinate of the circle's center must be equivalent to the length of the circle's radius. In other words, if the graph of $(x - h)^2 + (y - k)^2 = r^2$ is a circle that intersects the y -axis at exactly one point, then $r = |h|$ must be true. The equation in choice C is $(x - 4)^2 + (y - 9)^2 = 16$, or $(x - 4)^2 + (y - 9)^2 = 4^2$. This equation is in the form $(x - h)^2 + (y - k)^2 = r^2$, where $h = 4$, $k = 9$, and $r = 4$, and represents a circle in the xy -plane with center $(4, 9)$ and radius of length 4. Substituting 4 for r and 4 for h in the equation $r = |h|$ yields $4 = |4|$, or $4 = 4$, which is true. Therefore, the equation in choice C represents a circle in the xy -plane that intersects the y -axis at exactly one point.

Choice A is incorrect. This is the equation of a circle that does not intersect the y -axis at any point.

Choice B is incorrect. This is an equation of a circle that intersects the x -axis, not the y -axis, at exactly one point.

Choice D is incorrect. This is the equation of a circle with the center located on the y -axis and thus intersects the y -axis at exactly two points, not exactly one point.